Tema 2.1

1.

a_1	a_0	b_1	b_0	c_2	c_1	c_0
0	0	0	0	0	0	0
0	0	0	1	0	0	1
0	0	1	0	0	1	0
0	0	1	1	0	1	1
0	1	0	0	0	0	1
0	1	0	1	0	1	0
0	1	1	0	0	1	1
0	1	1	1	1	0	0
1	0	0	0	0	1	0
1	0	0	1	0	1	1
1	0	1	0	1	0	0
1	0	1	1	1	0	1
1	1	0	0	0	1	1
1	1	0	1	1	0	0
1	1	1	0	1	0	1
1	1	1	1	1	1	0

a_1a_0 \ b_1b_0	00	01	11	10
00	0	1	1	0
01	1	0	0	1
11	1	0	0	1
10	0	1	1	0

c_0

$$c_0 = \bar{a}_0 \cdot b_0 + a_0 \cdot \bar{b}_0$$

a_1a_0 \ b_1b_0	00	01	11	10
00	0	0	1	1
01	0	1	0	1
11	1	0	1	0
10	1	1	0	0

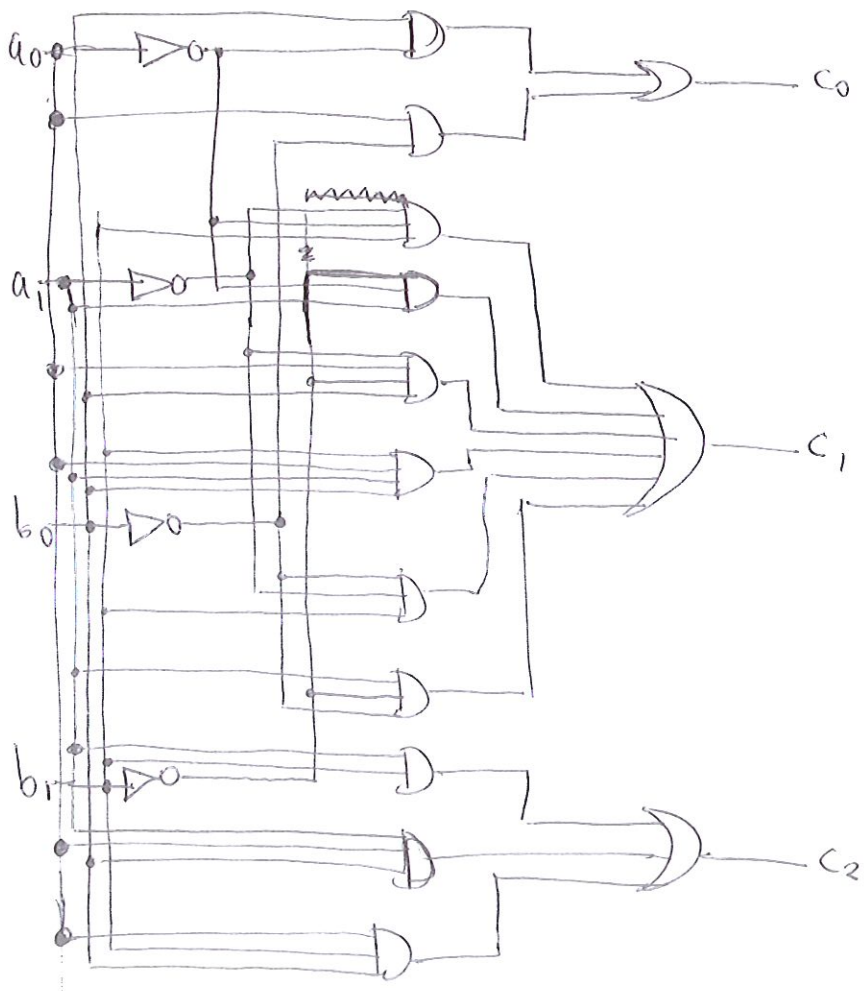
c_1

$$c_1 = \bar{a}_1 \cdot \bar{a}_0 \cdot b_1 + a_1 \cdot \bar{a}_0 \cdot \bar{b}_1 + \bar{a}_1 \cdot b_1 \cdot \bar{b}_0 + a_1 \cdot \bar{b}_1 \cdot \bar{b}_0 + \bar{a}_1 \cdot a_0 \cdot \bar{b}_1 \cdot b_0 + a_1 \cdot a_0 \cdot b_1 \cdot b_0$$

a_1a_0 \ b_1b_0	00	01	11	10
00	0	0	0	0
01	0	0	1	0
11	0	1	1	1
10	0	0	1	1

c_2

$$c_2 = a_1 \cdot b_1 + a_1 \cdot a_0 \cdot b_0 + a_0 \cdot b_1 \cdot b_0$$



2.

	OP1	OP0	A	B	Res	Desb.
sumar	0	0	0	0	0	0
	0	0	0	1	1	0
	0	0	1	0	1	0
	0	0	1	1	0	1
restar	0	1	0	0	0	0
	0	1	0	1	1	1
	0	1	1	0	1	0
	0	1	1	1	0	0
AND	1	0	0	0	0	0
	1	0	0	1	0	0
	1	0	1	0	0	0
	1	0	1	1	1	0
OR	1	1	0	0	0	0
	1	1	0	1	1	0
	1	1	1	0	1	0
	1	1	1	1	1	0

Res

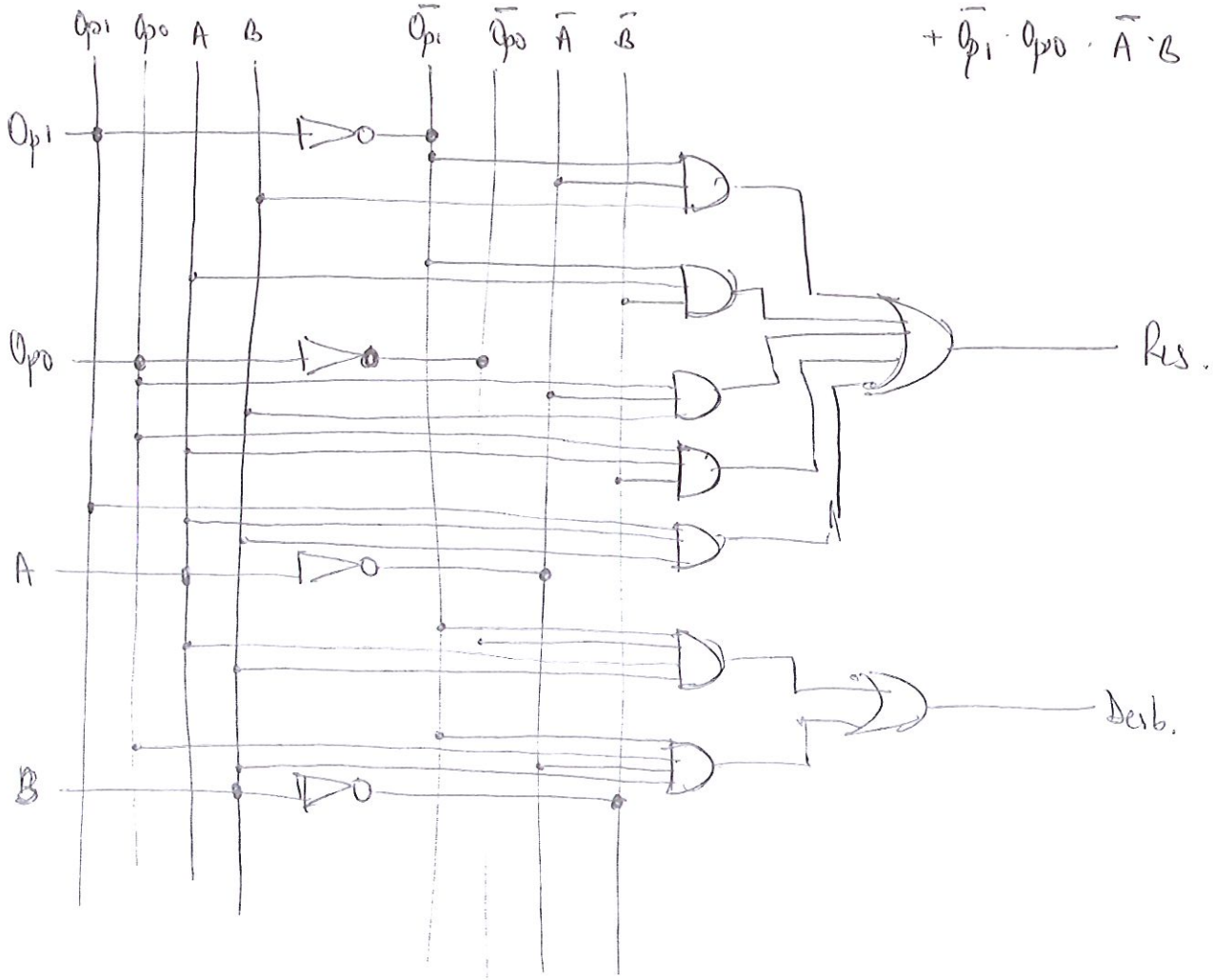
OP \ AB	00	01	11	10
00	0	1	0	1
01	0	1	0	1
11	0	1	1	1
10	0	0	1	0

$$Res = \overline{Op1} \cdot \bar{A} \cdot B + \overline{Op1} \cdot A \cdot \bar{B} + Op0 \cdot \bar{A} \cdot B + Op0 \cdot A \cdot \bar{B} + Op1 \cdot A \cdot B$$

Desb.

OP \ AB	00	01	11	10
00	0	0	1	0
01	0	1	0	0
11	0	0	0	0
10	0	0	0	0

$$Desb = \overline{Op1} \cdot \overline{Op0} \cdot A \cdot B + \overline{Op1} \cdot Op0 \cdot \bar{A} \cdot B$$



3. Las salidas A,B...G se ponen a 1 si el segmento correspondiente se tiene que iluminar.

a_3	a_2	a_1	a_0	A	B	C	D	E	F	G
0	0	0	0	1	1	1	1	1	1	0
0	0	0	1	0	1	1	0	0	0	0
0	0	1	0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1	0	0	1
0	1	0	0	0	1	1	0	0	1	1
0	1	0	1	1	0	1	1	0	1	1
0	1	1	0	1	0	1	1	1	1	1
0	1	1	1	1	1	1	0	0	0	0
1	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	1	1	0	0	1	1
1	0	1	0	1	1	1	0	1	1	1
1	0	1	1	0	0	1	1	1	1	1
1	1	0	0	1	0	0	1	1	1	0
1	1	0	1	0	1	1	1	1	0	1
1	1	1	0	1	0	0	1	1	1	1
1	1	1	1	1	0	0	0	1	1	1

En todos los casos hay menos 0s. Conviene implementar lógica negativa.

A

a_3a_2 \ a_1a_0	00	01	11	10
00	1	0	1	1
01	0	1	1	1
11	1	0	1	1
10	1	1	0	1

B

a_3a_2 \ a_1a_0	00	01	11	10
00	1	1	1	1
01	1	0	1	0
11	0	1	0	0
10	1	1	0	1

C

a_3a_2 \ a_1a_0	00	01	11	10
00	1	1	1	0
01	1	1	1	1
11	0	1	0	0
10	1	1	1	1

D

a_3a_2 \ a_1a_0	00	01	11	10
00	1	0	1	1
01	0	1	0	1
11	1	1	0	1
10	1	0	1	0

$$\bar{A} = \bar{a}_3 \cdot \bar{a}_2 \cdot \bar{a}_1 \cdot a_0 + \bar{a}_3 \cdot a_2 \cdot \bar{a}_1 \cdot \bar{a}_0 + a_3 \cdot a_2 \cdot \bar{a}_1 \cdot a_0 + a_3 \cdot \bar{a}_2 \cdot a_1 \cdot a_0$$

$$A = (a_3 + a_2 + a_1 + \bar{a}_0) \cdot (a_3 + \bar{a}_2 + a_1 + a_0) \cdot (\bar{a}_3 + \bar{a}_2 + a_1 + \bar{a}_0) \cdot (\bar{a}_3 + a_2 + \bar{a}_1 + \bar{a}_0)$$

$$\bar{B} = \bar{a}_3 \cdot a_2 \cdot \bar{a}_1 \cdot a_0 + a_3 \cdot a_2 \cdot \bar{a}_0 + a_2 \cdot a_1 \cdot \bar{a}_0 + a_3 \cdot a_1 \cdot a_0$$

$$B = (a_3 + \bar{a}_2 + a_1 + \bar{a}_0) \cdot (\bar{a}_3 + \bar{a}_2 + a_0) \cdot (\bar{a}_2 + \bar{a}_1 + \bar{a}_0) \cdot (\bar{a}_3 + \bar{a}_1 + \bar{a}_0)$$

$$\bar{C} = \bar{a}_3 \cdot \bar{a}_2 \cdot a_1 \cdot \bar{a}_0 + a_3 \cdot a_2 \cdot \bar{a}_0 + a_3 \cdot a_2 \cdot a_1$$

$$C = (a_3 + a_2 + \bar{a}_1 + a_0) \cdot (\bar{a}_3 + \bar{a}_2 + a_0) \cdot (\bar{a}_3 + \bar{a}_2 + \bar{a}_1)$$

$$\bar{D} = \bar{a}_3 \cdot a_2 \cdot \bar{a}_1 \cdot \bar{a}_0 + a_3 \cdot \bar{a}_2 \cdot a_1 \cdot \bar{a}_0 + \bar{a}_2 \cdot \bar{a}_1 \cdot a_0 + a_2 \cdot a_1 \cdot a_0$$

$$D = (a_3 + \bar{a}_2 + a_1 + a_0) \cdot (\bar{a}_3 + a_2 + \bar{a}_1 + a_0) \cdot (a_2 + a_1 + \bar{a}_0) \cdot (\bar{a}_2 + \bar{a}_1 + \bar{a}_0)$$

$a_3a_2 \backslash a_1a_0$	00	01	11	10
00	1	0	0	1
01	0	0	0	1
11	1	1	1	1
10	1	0	1	1

(E)

$a_3a_2 \backslash a_1a_0$	00	01	11	10
00	1	0	0	0
01	1	1	0	1
11	1	0	1	1
10	1	1	1	1

(F)

$a_3a_2 \backslash a_1a_0$	00	01	11	10
00	0	0	1	1
01	1	1	0	1
11	0	1	1	1
10	1	1	1	1

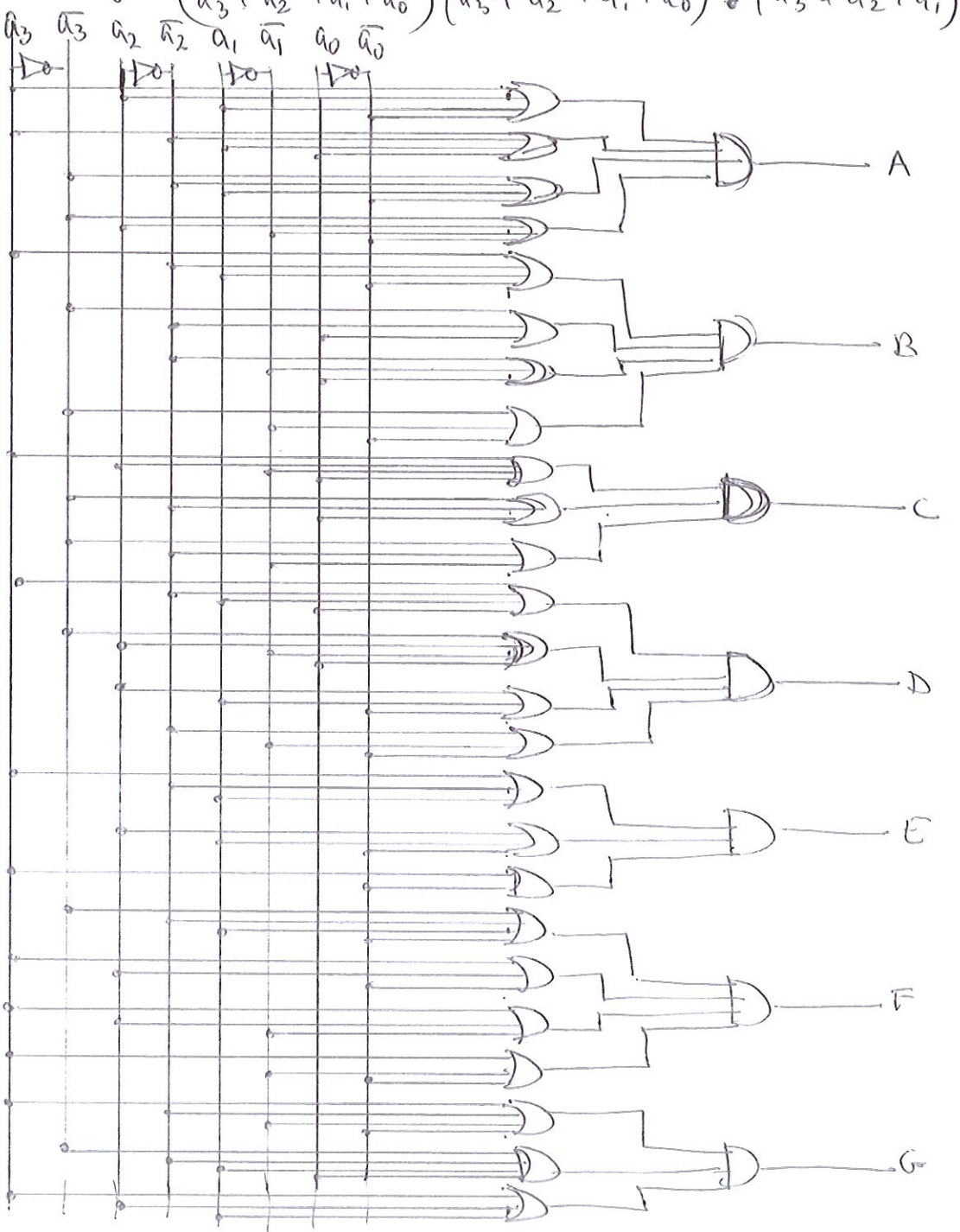
(G)

$$\overline{E} = \overline{a_3} \cdot a_2 \cdot \overline{a_1} + \overline{a_2} \cdot \overline{a_1} \cdot a_0 + \overline{a_3} \cdot a_0$$

$$E = \left(a_3 + \overline{a_2} + a_1\right) \cdot \left(a_2 + a_1 + \overline{a_0}\right) + \left(a_3 + \overline{a_0}\right)$$

$$\overline{F} = a_3 \cdot a_2 \cdot \overline{a_1} \cdot a_0 + \overline{a_3} \cdot \overline{a_2} \cdot a_0 + \overline{a_3} \cdot \overline{a_2} \cdot a_1 + \overline{a_3} \cdot a_1 \cdot a_0$$

$$F = \left(\overline{a_3} + \overline{a_2} + a_1 + \overline{a_0}\right) \cdot \left(a_3 + a_2 + \overline{a_0}\right) \cdot \left(a_3 + a_2 + \overline{a_1}\right) \cdot \left(a_3 + \overline{a_1} + \overline{a_0}\right)$$

$$\overline{G} = \overline{a_3} \cdot a_2 \cdot a_1 \cdot a_0 + a_3 \cdot a_2 \cdot \overline{a_1} \cdot \overline{a_0} + \overline{a_3} \cdot \overline{a_2} \cdot \overline{a_1}$$

$$G = \left(a_3 + \overline{a_2} + \overline{a_1} + \overline{a_0}\right) \cdot \left(\overline{a_3} + \overline{a_2} + a_1 + a_0\right) \cdot \left(\overline{a_3} + a_2 + a_1\right)$$